



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/984095

Date:

30/03/2023(16:45:00)

Subject: Review of LMI "GUIDELINES FOR CARRYING OUT SAFE ERECTION OF SCAFFOLDING INSIDE BOILER"

REF.: SMPP/O&M/MM/PP/

Date: 30.03.2023

Subject: Review of LMI "GUIDELINES FOR CARRYING OUT SAFE ERECTION OF SCAFFOLDING INSIDE BOILER"

LMI No: LMI/OGN/OPS/MECH/046/

The above approved LMI was prepared based on reference document - " **COS-ISO-00-OGN/OPS/MECH/046**" circulated by CC-OS. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

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On: 30/03/2023(16:45:00) **Ref:** SMTTP/MECH MAINT-/2022-23/LMI/984095

The changes have been incorporated in the LMI "Guidelines for carrying out safe erection of scaffolding inside boiler" as per changes mentioned in the reference document " COS-ISO-00-OGN/OPS/MECH/046".

Competent authority may please accord approval for this revised LMI with revision no-1 of document no-"LMI/OGN/OPS/MECH/046".

Submitted by:

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On:01/04/2023(17:06:19) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/984095

Following changes incorporated in revised LMI (attached as annexure "NEW LMI"):

1. Clause 7 (A)- Points 10,11,12 added.
2. Clause 7 (C)- Points 10,11,12,13 added
3. Clause 7 (F)- Points 9,10 added
4. Changes incorporated in Annexure-D as per OGN

Submitted for approval please.

Submitted by:

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On:03/04/2023(11:29:49) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/984095

Revised LMI may please be accorded approval.

Submitted by:

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Approved & Submitted by:

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On:04/04/2023(17:26:50) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/984095

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/046

ISSUE NO.1; REVISION NO. 1.00; DATE: MAR'2023

**GUIDELINES FOR CARRYING OUT SAFE
ERECTION OF SCAFFOLDING INSIDE
BOILER**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	GUIDELINES FOR CARRYING OUT SAFE ERECTION OF SCAFFOLDING INSIDE BOILER
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/046/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV: 1; MAR'2023</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/MECH/046/ Rev 1</u>
<u>4.PREPARED BY :</u> SR. MANAGER (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY :</u> DGM (MM-PP)	SIGNATURE-----
AGM (BMD)	SIGNATURE-----
AGM (EEMG)	SIGNATURE-----
GM (MTC)	SIGNATURE-----
GM (O & M)	SIGNATURE-----
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE:----- HOP, SIMHADRI
<u>7.DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.

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<u>2.0 INTRODUCTION</u> Different types of scaffolding are being regularly used for inspection and repair of Boiler pressure parts. Safety is the most important aspect while using the scaffolding system inside boiler during overhauls/breakdowns/planned maintenance. Hence it is necessary that each person who intends to utilize the scaffolding system take the time to learn the proper and safe methods relating to erection and commissioning. Safety begins by familiarizing with the functions and limitations of the scaffolding system. The objective of this LMI is to improve the safety of scaffolding erected inside boiler by following specific guidelines. <u>3.0 SCOPE</u> The scope of this LMI is to detail the practices to be followed while using scaffold materials to access different locations of boiler like Reheater, Platen SH, Water Wall or any hanging coils in boiler furnace, goose neck area or second pass, roof tubes for maintenance, inspection or any other job requiring access at height. <u>4.0 REFERENCE DOCUMENT</u> Operation Guidance note: COS-ISO-00-OGN/OPS/MECH/046 Rev.1, circulated by operation services - NTPC Corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> <u>COS-ISO-00-OGN/OPS/MECH/046/ Rev 0</u> <u>6.0 SPECIFIC INSTRUCTIONS</u> The objective of the LMI is sought to be achieved through following actions: a) By carrying out specific checks and instructions at different stages of using scaffolding inside boiler i.e., during erection, before using the platforms, while using the platforms, during dismantling etc. b) By carrying out regular maintenance of scaffolding equipment		

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<u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u>		
<u>A) PERSONNEL SAFETY DURING SCAFFOLDING ERECTION</u>		
1. All the competent persons who are involved in erection and dismantling of scaffolding must fully comprehend and act in accordance with OEM / OES manual and drawings. 2. A competent person is a designated person, suitably trained by OEM/OES representative, qualified by knowledge and practical experience to carry out erection and dismantling of scaffolding system. 3. Competent persons must be emotionally strong and physically able to withstand the stress of working at high elevations. 4. All Workers involved in scaffolding work should compulsorily undergo Vertigo test and other health checkup to ensure that they are physically fit for carrying out scaffolding work. 5. All persons engaged in erection and dismantling of scaffolding system must always wear PPE's like (full body harness) safety belts, safety shoes, Helmet, Goggles etc. 6. Maintain communication with the supervisor at ground level continuously. 7. Make the copy of OEM / OES manual & drawings available to all concerned supervisors/Area Engineers. 8. During Boiler Overhauling, OEM/OES personnel or his representative shall be called for Supervision & Inspection of scaffolding erection. 9. Fall arrestor should be used while working at height. 10. Aluminum scaffolding material such as span decks, snap outs etc. should be lifted and lowered with the portable winches of suitable capacity. 11. Portable winches rope and electrical motor to be tested before use with valid test certificates. 12. Walkie Talkie/suitable communication devices to be used for immediate communication.		
Responsibility - HOD (BMD)		
<u>B) CHECK POINTS BEFORE USE OF SCAFFOLDING SYSTEM</u>		
1. The Scaffolding should be stable, strong, and safe to use. 2. Inspect the Scaffolding before use, erection, re-erection and after dismantling and at regular intervals to make sure that it is maintained and in safe workable condition. 3. The EIC should assign a competent person for inspection purpose. Competent Person must be trained and certified for scaffolding erection work.		

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4. Availability of OEM/OES guidelines, drawings & calculation of safe working load required for working on scaffold. 5. Check physical condition of platforms and other members. It should not be damaged. All damaged components should be marked, recorded, and removed from the lot to be used. 6. Scaffolding clamps to be used for fastening members. 7. Determining appropriate control measures for risk in consultation with all relevant persons involved in the work, including the principal contractor, scaffolding contractor, designers, plant operators, safety personnel etc. 8. Check for adequate light inside the boiler/area concerned. 9. Check bottom ash hoppers and eco hoppers are sealed properly. 10. Check for any dangerously hanging clinker/soot 11. A pre-start or pre-use check maybe conducted by adequately trained / experienced person/persons (Safety & User department) before a scaffold in use is first used for the day. This must include a visual check. The scaffold must still be checked daily till in use. Responsibility - HOD (BMD) & Scaffolding and Ladder Inspector <u>C) CHECK POINTS DURING SCAFFOLDING ERECTION</u> 1. Emergency Rescue arrangement should be in place before start of scaffolding erection. 2. Availability of First Aid facility/ Ambulance at site. 3. Area should be properly lighted and ventilated. 4. All persons involved in work should always wear PPE. 5. All footings should be sound, rigidly set and leveled by plumb/sprit level/water level. 6. Scaffolding should be able to bear two times the maximum intended load. 7. Double layered Safety net must be used below the scaffolding during and after erection. The Double layered safety net should be installed at every 6 meter height in the scaffolding. 8. Periodical inspection by competent persons for proper scaffolding erection with respect to drawing, safety, and rigidity of structure. 9. Welding of scaffolding base of cup lock with boiler tube fins to be done only. 10. DPT of fins and boiler tubes to be done after cutting scaffolding welded base. Thickness of tube hidden below scaffolding base to be done. 11. Double layered safety net to be installed after taking support from boiler water wall as per OEM drawing. 12. Subsequent scaffolding to be done after complete erection of previous levels. 13. Coir rope should not be used for making ladder step in scaffolding.		

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14. Only healthy/standard components of scaffolding system should be used. 15. All damaged components should be marked, recorded, and removed from site. 16. For making any working/ load bearing platform, only heavy-duty scaffolding pipes should be used, medium duty pipes may be used for handrails, bracing and strutting. 17. In case of round the clock working, workers should change every shift. 18. Scaffolding should be provided with safe means of access such as stairs, ladders, and ramps. Bracings should not be used as a means of access. 19. Guard rail and toe boards shall be fitted in position. Guard rail/ handrails must be provided at 1 to 1.1 mtr height from the platform and all guard rails shall be provided with mid rails. 20. Provide Toe Guards on all platforms. 21. Avoid overloading. The load on shaft (load bearing element) should be evenly distributed as far as practicable. 22. Do not use ladders and makeshift devices on top of scaffold to increase height. 23. Avoid accumulation of unwanted material on scaffolding platform. The scaffolding platform shall be cleaned regularly. 24. After Water Washing and HP Jet cleaning of Boiler, scaffolding is cleaned and free from debris. 25. Welding is not allowed in scaffolding except at footings (base plate) which shall be welded with S Panel Fins. 26. Minimum gap shall be maintained between water wall and scaffold. 27. Scaffold should be used only after complete erection and certification by competent person/ OEM or OES representative / Safety personnel. Note: In case of any problem at height ask for help of competent persons. Do not try to modify/rectify/alter it yourself. Responsibility - HOD (BMD) & Scaffolding and Ladder Inspector		

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D) INSPECTION REPORTS

1. It is recommended that all scaffolds should be inspected using the appropriate inspection report to show that best practices and minimum requirements have been met. 3 to 4 persons in the project/station should be identified as competent person, who should be trained and certified. The competent person or area engineer along with the safety officer and OEM/OES representative should jointly check the scaffolding and certify the same as "Safe to use" after all compliances.
2. TAGGING: Safe to use scaffold tags should be developed which should be green in color and put at the entry of the scaffolding once it is certified for use. Similarly, for any scaffolding under erection and not safe to use should be tagged as Not fit for use with red tag at the entry point. Also, its entry point should be barricaded preventing any unintended / unsafe use. The inspection reports can also be used when checking scaffolds for their register checks.

Responsibility - HOD (BMD) & Scaffolding and Ladder Inspector

E) SCAFFOLD REGISTER/INSPECTION RECORD REQUIREMENTS

1. For scaffolding erected in Boiler, a scaffold register be kept on site and be available for inspection along with drawing for inspection.
2. When complete, the inspection records should show:
 - a) Project name, address, or other clear, identification.
 - b) The location of the scaffold with respect to site coordinates or the location at the Boiler or structure, so that the scaffolding can be clearly located.
 - c) Miscellaneous details of the scaffold.
 - d) A record of each inspection carried out.
3. The inspections must be carried out at the following intervals:
 - a) Before the scaffold is first put to in to use.
 - b) Daily, while the scaffold is in use.
 - c) After any structural alteration or addition.
 - d) Weekly inspection when the scaffold is setup but not in use.
4. The inspection must be carried out by either:
 - a) A certificated scaffolder of the appropriate class; or
 - b) A competent person/ OEM or OES Representative/ Safety Personnel
5. The entries of each inspection in the scaffold register must be made and signed by the person who carried out the inspection.

Responsibility - HOD (BMD) & Scaffolding and Ladder Inspector

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F) MAINTENANCE OF SCAFFOLDING EQUIPMENT

1. Never drop any scaffolding material from height.
2. Scaffolding members used in Boiler should be free from rust. Painting of all scaffolding material to be done to keep it corrosion free.
3. Only Heavy Grade scaffolding material to be procured for Boiler scaffolding.
4. New Scaffolding materials shall have long lasting rust preventive coatings like galvanizing components etc. Aluminum scaffolding may be preferred during procurement.
5. 5% of additional scaffolding members shall be procured or kept for replacement of damaged components.
6. Different color codes may be used for different types of scaffolding items.
7. Store scaffolding material when not in use in a shade /covered area and stack on raised platform so that stored material does not get affected by rain/weather.
8. Scaffold accessories such as coupler, joint pins, spigots etc. must be cleaned and greased at nut/bolts after every use.
9. **Damaged aluminum scaffolding materials to be replaced with new material with proper record of damaged aluminum scaffolding materials.**
10. **Scaffolding material to be cleaned before and after use in boiler.**

Responsibility - HOD (BMD)

8.0 RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to Safe Erection of Scaffolding inside Boiler as detailed in the LMI are with HOD (BMD) & HOD (SAFETY)

9.0 RECORD KEEPING

The records to be maintained under this LMI include:

1. Compliance record of the activities detailed under this LMI
2. A record of the trained persons should be kept at site as per Annexure A
3. Protocol as per Annexure B signed by BMD and Safety department before use of Scaffolding

The above records are to be maintained by HOD (BMD).

10.0 REVIEW

This document shall be reviewed by Head of The Project on two yearly basis.

[illegible]

[illegible]

NTPC	13.0 COMPLIANCE RECORD	SIMHADRI
TITLE: GUIDELINES FOR CARRYING OUT SAFE ERECTION OF SCAFFOLDING INSIDE BOILER REV: 1 DATE: MAR'23		
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UNIT NO.	ACTIVITY	COMPLIANCE

[illegible]

EIC

OEM/OES Representative

NTPC	ANNEXURE-B	SIMHADRI
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<u>Protocol for use of Scaffolding</u> Unit No: _____ Date: _____ Reason For Unit Shutdown: _____ The following protocol should be signed by BMD and Safety department before use of Sky Climber : All areas of scaffolding are checked carefully and found that all requirements of safe scaffolding are met. The erected scaffolding is Stable, Strong, and Safe to use. The scaffolding is tagged "Safe to use Scaffold"		

(AGENCY)

(BMD)

(Scaffolding and Ladder Inspector)

(OEM/OES)

NTPC		ANNEXURE-C	SIMHADRI
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<u>Scaffolding Register/ Inspection record</u>			
Sl. No	Description	Remark	
1	Project/station detail		
2	Date /Location of scaffolding erected		
3	Reference of Drawings/sketch		
4	Safe working load intended		
5	Type of inspection (First use, daily use, after alteration etc.)		
6	Date /Inspection report/Remark		
7	Check physical condition of platform and other members	As per check list as Annexure-D	
8	Name and Sign of Inspecting authority		

NTPC		ANNEXURE-D		SIMHADRI	
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<u>CHECKLIST</u>					
Sl. No	Activity Description			Status	Remarks
1	Adequacy of scaffoldings & Safe accessibility arrangements.			YES/NO/NA	
2	Suitability & soundness of material used for supports & scaffoldings.			YES/NO/NA	
3	Adequacy of Bracing & strutting to ensure stability			YES/NO/NA	
4	Adequacy of Fastening to prevent toppling.			YES/NO/NA	
5	Walkie Talkie/ Suitable communication devices available with all concerned worker/ Supervisors			YES/NO/NA	
6	OEM/OES personnel or his representative for scaffolding available for scaffolding erection (QES) during boiler overhauling			YES/NO/NA	
7	Proximity of platform at working place			YES/NO/NA	
8	Ensure daily routine Checks &record keeping of observations			YES/NO/NA	
9	Check for any dangerously hanging clinker/soots inside Boiler			YES/NO/NA	
10	No Coir rope used for making ladder steps for moving on scaffolding			YES/NO/NA	
11	Whether any moving/ rotating part is rubbing against the scaffolding?			YES/NO/NA	
12	Is access & Way out to/from working Platforms is free from obstructions?			YES/NO/NA	
13	Removal of unwanted loose material from the working platforms.			YES/NO/NA	
14	Check for loose or hanging members			YES/NO/NA	
15	Check for Guard Rails and Toe Boards			YES/NO/NA	
16	All working platforms are adequately Guarded by providing handrails & toe guards			YES/NO/NA	
17	Sagging/ deformed/ damaged planks replaced immediately from scaffolding				
18	Any joints in the scaffolding arrangement are being welded? If yes, ensure full welding (Stitch welding not allowed)			YES/NO/NA	
19	Ensure that locking joints are pinned at each joint.			YES/NO/NA	
20	Check all staircases& objects being erected for overhead projections.			YES/NO/NA	
21	Check for withdrawal of persons from heights during storms or high winds			YES/NO/NA	

22	Whether all footings of scaffoldings are firmly positioned on preferably steel base	YES/NO/NA	
18	OEM safety guidelines, drawings, Maximum allowable persons/load limit displayed	YES/NO/NA	
19	Provision of fall arrestors /safety nets	YES/NO/NA	
20	Label/ Display of safe to work	YES/NO/NA	
21	Winches used in scaffolding have validate test certificate	YES/NO/NA	
22	Aluminum scaffolding cleaned before erection	YES/NO/NA	
23	Aluminum scaffolding cleaned after dismantling	YES/NO/NA	
24	Welding of all cup lock scaffolding base is welded to boiler tube fin	YES/NO/NA	
25	DPT of all fins and adjacent boiler tubes done in S Panel after cutting of cup lock scaffolding base	YES/NO/NA	
26	After dismantling and removal of scaffolding base, thickness of boiler tube hidden below scaffolding base to be checked.	YES/NO/NA	

(AGENCY)

(BMD)

(Scaffolding and Ladder Inspector)

(OEM/OES)